The Fine-Structure Constant as a Size Ratio

The Hydrogen Atom and the Deuteron in One Coulomb Picture

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Mathematical theory and formulation by the author; manuscript developed in collaboration with Claude (Anthropic). See Section 8.

Abstract

The hydrogen atom is a proton with an electron shell of size a_0 (the Bohr radius). The deuteron — the ^{2}H nucleus — is, in a pure proton–electron picture (neutron = proton + electron), the system $\mathbf{p} + \mathbf{e} + \mathbf{p}$: two protons bound by one shared electron, of size R_p (a few fm). The two are the *same* Coulomb physics — protons and one electron — differing only in the electron's **size**: bound by a single proton's *charge* it relaxes to a_0; caged between two protons it compacts to the proton *size* R_p. The nuclear/atomic size ratio is then the fine-structure constant squared,

$$\alpha^2 = \frac{R_p}{a_0} = \frac{\text{proton (deuteron) size}}{\text{atomic size}},$$

so α is the **proton-to-atom size ratio** — a pure ratio of two lengths, **independent of the mass m and of the speed of light c**, not a coupling constant. The relation is bracketed by two independent quantitative validations of the same program: atomic total energies $< 1\%$ across the periodic table (a_0 side), and the He-4/deuteron binding-energy ratio 13.1 vs. 12.7 ($\sim 3\%$, R_p side). Mass plays no role in this electrostatic structure; it is a separate, inertial/gravitational quantity. Built from charge and a kinetic coefficient alone, no strong force.

Status. A structural/interpretive proposal within the RealQM/RealNucleus program. The claim is the relation $\alpha^2 = R_p/a_0$ read off the H atom and the deuteron; the underlying primitive (the proton's finite size, or the classical radius) is stated as open (Section 6). A cosmological extension is noted only briefly (Section 7).

Keywords: fine-structure constant; hydrogen atom; deuteron; Coulomb binding; classical electron radius; proton radius; RealQM; RealNucleus.

1 The question

Why is the nucleus roughly 10^4–10^5 times smaller than the atom that contains it? In standard physics the atomic scale is electromagnetic (set by $\hbar, m, e$) and the nuclear scale is set by the strong force — two unrelated sectors — and $\alpha = e^2/\hbar c$ enters as the coupling of the electromagnetic interaction. We point out that the simplest atom and the simplest composite nucleus — the **hydrogen atom** and the **deuteron** — are, in a pure proton–electron (Coulomb) picture, the *same* system at two sizes, and that their size ratio *is* α^2. The argument is elementary and uses no relativity.

2 Two systems, one physics: the H atom and the deuteron

An electron is a normalised charge cloud ψ ($\int \psi^2 = 1$) with energy $E = \kappa\!\int |\nabla\psi|^2 - Z\,e^2/L$, where $\kappa = \hbar^2/2m$ is a kinetic coefficient. The whole structure below needs only three inputs: the charge e^2, the coefficient κ, and the proton's finite size R_p — *no mass, no strong force, no c.*

Hydrogen atom: p + e, size a_0. A single proton with one electron. Attracted to the proton's charge, the electron must *taper to zero* in the surrounding vacuum; that varying profile costs kinetic energy $\sim\kappa/L^2$, and minimising $\kappa/L^2 - e^2/L$ gives

$$L = a_0 = \frac{2\kappa}{e^2} = \frac{\hbar^2}{me^2} \approx 0.53\ \text{Å},$$

the Bohr radius — the electron shell size *set by the proton's charge.* To this electron the proton's finite size is irrelevant (it is 10^4–10^5 times smaller, effectively a point).

Deuteron: p + e + p, size R_p. The deuteron is the ^{2}H nucleus, a proton plus a neutron. In the pure proton–electron model (neutron = p + e [2]) the deuteron is therefore

$$\mathrm{d} = \mathrm{p} + \mathrm{n} = \mathrm{p} + (\mathrm{p} + \mathrm{e}) = \mathbf{p{+}e{+}p},$$

two protons bound by one shared electron — the same constituents as the H atom, plus a second proton; in fact the nuclear analogue of the molecular ion H_2^+. Now the electron is *caged* between the protons on its own domain with a multiphase *free boundary*: its density does *not* drop to zero but hands off to the proton density. A flat caged density then has $\nabla\psi = 0$, hence *zero kinetic energy* — there is no kinetic penalty for being small. With no kinetic pressure to spread it, the electron's size is set by the *proton's finite size* R_p (the cage floor that stops collapse), and its binding is Coulomb at that size (∼MeV at fm). The electron's size is now *set by the proton's size.*

Same physics, two sizes. The two systems are pure proton–electron Coulomb; the electron simply has no fixed size. The distinction is **proton charge vs. proton size**:

	constituents	electron size	binding
H atom	p + e	$a_0 \approx 0.5$ Å (charge-set)	∼13.6 eV
deuteron ^{2}H	p + e + p	$R_p \approx 2$ fm (size-set)	∼2.2 MeV

The deuteron's mirror twin: H$^-$. The deuteron has an exact charge conjugate. Swapping every + with − turns p + e + p (+ − +) into e + p + e (− + −) = **H$^-$**, a proton binding two electrons. The two are the *same 3-body system* — a central charge holding an outer like-charged pair against its own repulsion — differing in *nothing but* p ↔ e. So the entire size gap between them is the rigid-proton/fluid-electron asymmetry:

- deuteron: outer pair = **protons** (rigid, ∼fm) ⇒ size R_p (nuclear);
- H$^-$: outer pair = **electrons** (fluid, ∼Å) ⇒ size $\sim a_0$ (atomic),

and (deuteron size)/(H$^-$ size) $\approx R_p/a_0 = \alpha^2$. Charge is *mirrored, not changed* — so the α^2 gap is purely a *size* asymmetry, the sharpest statement of the thesis. (We keep the plain H atom, size exactly a_0, as the quantitative reference: H$^-$ is diffuse and weakly bound — radius $\sim 2.5\,a_0$, electron affinity only 0.75 eV — and an anion, so it gives the clean *symmetry* statement, not the clean *number*.)

3 α as a size ratio

Combining the two sizes, the deuteron/atom (nuclear/atomic) ratio is

$$\boxed{\alpha^2 = \frac{R_p}{a_0} = \frac{\text{deuteron (proton) size}}{\text{atomic size}}, \qquad \alpha = \sqrt{R_p/a_0}.} \tag{1}$$

This is the textbook relation $r_e/a_0 = \alpha^2$ in disguise. The electron has three lengths built from $(\hbar, m, c, e)$ — the Bohr radius $a_0 = \hbar^2/me^2$, the reduced Compton wavelength $\bar{\lambda}_C = \hbar/mc$, and the classical radius $r_e = e^2/mc^2$ — which, using $e^2 = \alpha\hbar c$, sit on the ladder

$$a_0 : \bar{\lambda}_C : r_e = 1 : \alpha : \alpha^2, \qquad \text{so} \qquad \frac{r_e}{a_0} = \alpha^2.$$

Our relation is this with the deuteron size identified as the classical radius, $R_p \approx r_e \approx 2.8$ fm. So α is reframed **geometrically** — the proton-to-atom size ratio, not a coupling constant — and the question "why $\alpha \approx 1/137$" becomes "*why is the deuteron* $\sim$*fm while the atom is* $\sim$*Å*": relocated to the proton size, an input.

α is independent of m and c. Written as a ratio of two lengths, $\alpha = \sqrt{R_p/a_0}$ is obtained by *measuring two sizes and dividing* — the deuteron and the atom. It therefore carries **no mass m and no speed of light c**: it is pure geometry, dimensionless by construction, fixed once the two lengths are given. The standard form $\alpha = e^2/\hbar c$ obscures this by writing the same number through c (and, via $a_0 = \hbar^2/me^2$, through m); but the ratio itself needs neither. In this program the primitive is the kinetic coefficient κ (so $a_0 = 2\kappa/e^2$), not the mass, and there is no c at all — the construction is non-relativistic and electrostatic. α is thus the most primitive of the three readings: a bare geometric fact about how much larger the atom is than the nucleus. (Standard $\alpha = e^2/\hbar c$ contains $\hbar$ and is intrinsically quantum; the only purely classical ingredient is $r_e = e^2/mc^2$, the radius at which the electron's electrostatic self-energy equals its rest energy — the electromagnetic-mass radius of Lorentz–Poincaré [4].)

α in standard quantum mechanics, for contrast. In standard QM/QED $\alpha = e^2/\hbar c$ is the central dimensionless constant of electromagnetism: the *coupling strength* (an amplitude $\sqrt{\alpha}$ at each photon vertex, so QED expands in powers of α); the scale of the *fine-structure* splitting it is named for ($\propto \alpha^2$, a relativistic spin–orbit correction); the Bohr-orbit ratio $v_1/c = \alpha$; and even a *running* coupling that grows with energy. Every one of these meanings is tied to c and $\hbar$ — there α is intrinsically relativistic and quantum. The reading here is the opposite: the *same number* as a bare **mass- and c-free length ratio** $\sqrt{R_p/a_0}$. The irony is that the constant named after a relativistic spectral effect is, at root, a piece of geometry — how much larger the atom is than the nucleus.

4 Empirical check

The size ratio. H atom: $a_0 = 0.53$ Å $\approx 53{,}000$ fm. Deuteron: rms radius ≈ 2.1 fm. Ratio d/H $\approx 1/25{,}000$, versus $\alpha^2 \approx 1/18{,}800$ — **agreement to** $\sim$30%, the deuteron sitting essentially at the classical electron radius (2.8 fm). The nuclear/atomic size ratio thus comes out as $1/\alpha^2$ from charge, $\hbar$ and the proton size alone, no strong force. (The deuteron is unusually loosely bound, so this is the scale/ratio being right, not a precise fit.)

The two scales are independently validated, bracketing the relation. Beyond the size ratio, the same Coulomb program is quantitative at *each* scale the relation connects:

- **Atomic side (a_0):** RealQM reproduces atomic **total energies to $< 1\%$ across the whole periodic table** (all $Z \geq 11$; the period-2 p-block, B–F, the only 3–6% exception) [3].
- **Nuclear side (R_p):** the same proton–electron Coulomb model gives the **He-4/deuteron binding-energy ratio** 13.1 **vs.** 12.7 **experimental** ($\sim 3\%$, nested charge-cloud shells) [2].

So $\alpha^2 = R_p/a_0$ sits between two independent quantitative validations — the atomic scale to $< 1\%$ and the nuclear binding ratio to $\sim$3% — not merely the $\sim$30% size check.

5 Mass is a separate quantity

Nothing above used mass: the H atom and the deuteron are built from charge and κ. Mass is **inertia** — the localisation (frequency / inverse size) of the matter wave, the dispersion gap $\omega_0 = mc^2/\hbar$ — a *separate, gravitational* quantity (signed, hence not the positive-definite electromagnetic field energy), coupling to the electrostatic sector only through energy, $E = mc^2$. Electromagnetism (charge) is an interaction; mass is the inverse size of each matter wave.

6 Complexity from the size asymmetry, and open problems

The asymmetry between proton and electron is not one of *charge* (charge is exactly balanced, $\pm e$) but of *size*: the proton is fixed-and-small (a rigid anchor), the electron variable. The fixed proton supplies anchors; the variable electron spans scales — a shell in the atom, a cage-filling cloud in the deuteron — producing the nested hierarchy that *is* complexity. With a fixed size for both, only one scale exists and nothing complex is built. And α measures this asymmetry: $\alpha^2 = R_p/a_0$ is the electron's inverse dynamic range ($1/\alpha^2 \approx 18{,}800$), so **the smallness of α is the engine of complexity**.

Open. (i) The nuclear **primitive**: the proton's finite size R_p as input length, or equivalently the relativistic classical radius $r_e = \alpha^2 a_0$ (electromagnetic mass, in the sense of Many-Minds Relativity [4]); α's value is reinterpreted, not derived. (ii) Whether the multiphase solver realises a flat, zero-kinetic-energy caged electron at the proton scale has not been cleanly confirmed (a symmetric proton cage is the proper test). (iii) Reconciling the rigid-proton/ fluid-electron picture with the measured proton/electron mass ratio.

7 Speculative outlook

The same ingredients extend, far more speculatively, to a cosmological narrative — equal-mass $\pm$ charges read off as the curvature of two fluctuating potentials, a gravitational collapse–bounce with negative-mass dark energy, and a primordial helium fraction $Y \approx 0.245$ — developed, with its (substantial) open problems up front, in a companion essay [5]; it is mentioned here only to indicate scope and is not part of the present, structural claim.

8 Collaboration

The mathematical theory and the proposal are the author's. The manuscript was developed in dialogue with Claude (Anthropic), which assisted in articulating the argument and surfacing its quantitative checks and open problems. We present the human–AI collaboration transparently as part of the working method.

References

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[3] C. Johnson, *RealQM total energies and ionization energies across the periodic table* (atomic total energies within 1% for $Z \geq 11$), manuscript, same repository.

[4] C. Johnson, *Many-Minds Relativity* — light as a medium wave ($c = \sqrt{T/\mu}$), mass as field self-energy ($P = mc \Rightarrow E = mc^2$), and a critique of Einstein's special relativity.

[5] C. Johnson, *Real Cosmology: a proton–electron universe from two fluctuating potentials* (companion essay), same repository.